January 2026

Thesis Presentation

Product Research Proposal
for AG Big Data Analytics

Discovering Impact: characterizing learning interaction data in educational contexts

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Agenda Overview

01 BACKGROUND OF THE STUDY

02 PROBLEM STATEMENT

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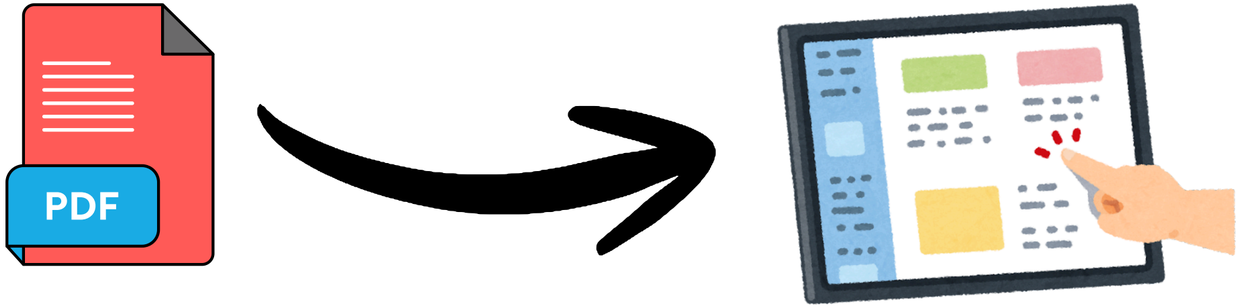
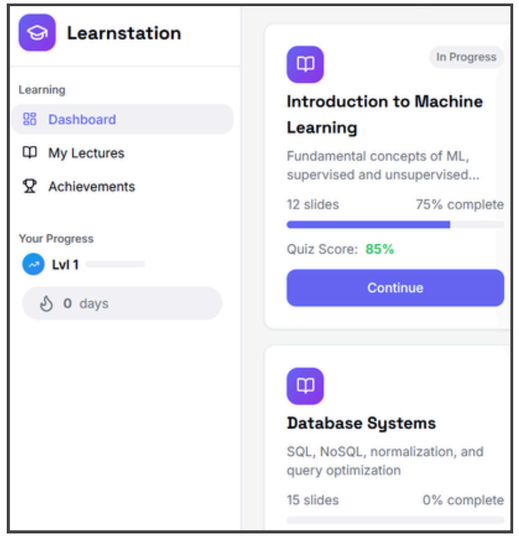
07 VISUALIZATION

08 WHAT MAKES OUR LEARNSTATION DIFFERENT?

Background of the study

Learnstation, an interactive learning platform that converts static lecture materials into interactive gamified learning experiences while collecting learner interaction data.





Problem Statement

01.

Students download slides and read linearly

02.

No immediate feedback on comprehension

03.

Absence of engagement incentives

Research Questions

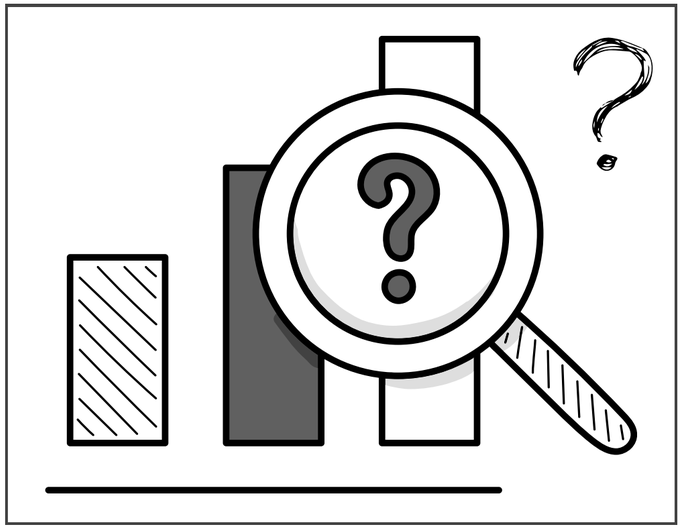
We want to addresse the following research questions:

RQ1: Which data profiling metrics are most suitable for characterizing learning interaction data in educational contexts?

RQ2: How do students perceive motivation and engagement in a gamified, data-driven learning platform?

RQ3: How can heterogeneous learning interaction events be efficiently aggregated, profiled, and analyzed to provide meaningful educational insights?

RQ*: Does interactive learning content improve learning efficiency and usability compared to static slide-based learning?



Untapped Data Potential

Modern LMS platforms generate substantial volumes of interaction data:

Access timestamps and frequency

Navigation patterns

Quiz attempts and scores

Session durations and Resource engagement metrics

Scientific Relevance

- **Event-Based Learning Analytics Framework**
- **Data Profiling Methodology**
- **Gamification Effectiveness Analysis**
- **LMS Integration Pattern**: Documented approach for external analytics platforms

Practical Contributions

- **Functional Prototype:** Deployable platform for the Database Systems course
- **Evaluation Dataset:** Anonymized interaction data for future research
- **Implementation Documentation:** Technical reference for similar projects

Data Profiling Application

Data Profiling Metrics

Category	Metrics
Engagement	Session frequency, duration distribution, return rate, streak consistency
Performance	Quiz accuracy, improvement trajectory, topic mastery, time-to-correct
Behavior	Navigation patterns, hint usage, revision frequency, content consumption order
Anomalies	Disengagement signals, unusual patterns, outlier detection

Example

Slide-Level Performance Analytics

Metric	What It Reveals
Quiz success rate per slide	Which concepts are poorly understood
Time spent per slide	Which slides are confusing or too dense
Retry frequency	Where students need multiple attempts
Skip rate	Which content students avoid

The Case for an Independent Platform

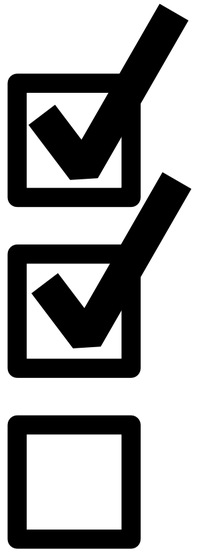
Direct ILIAS integration faces technical and institutional barriers.

Phase A: Standalone Platform

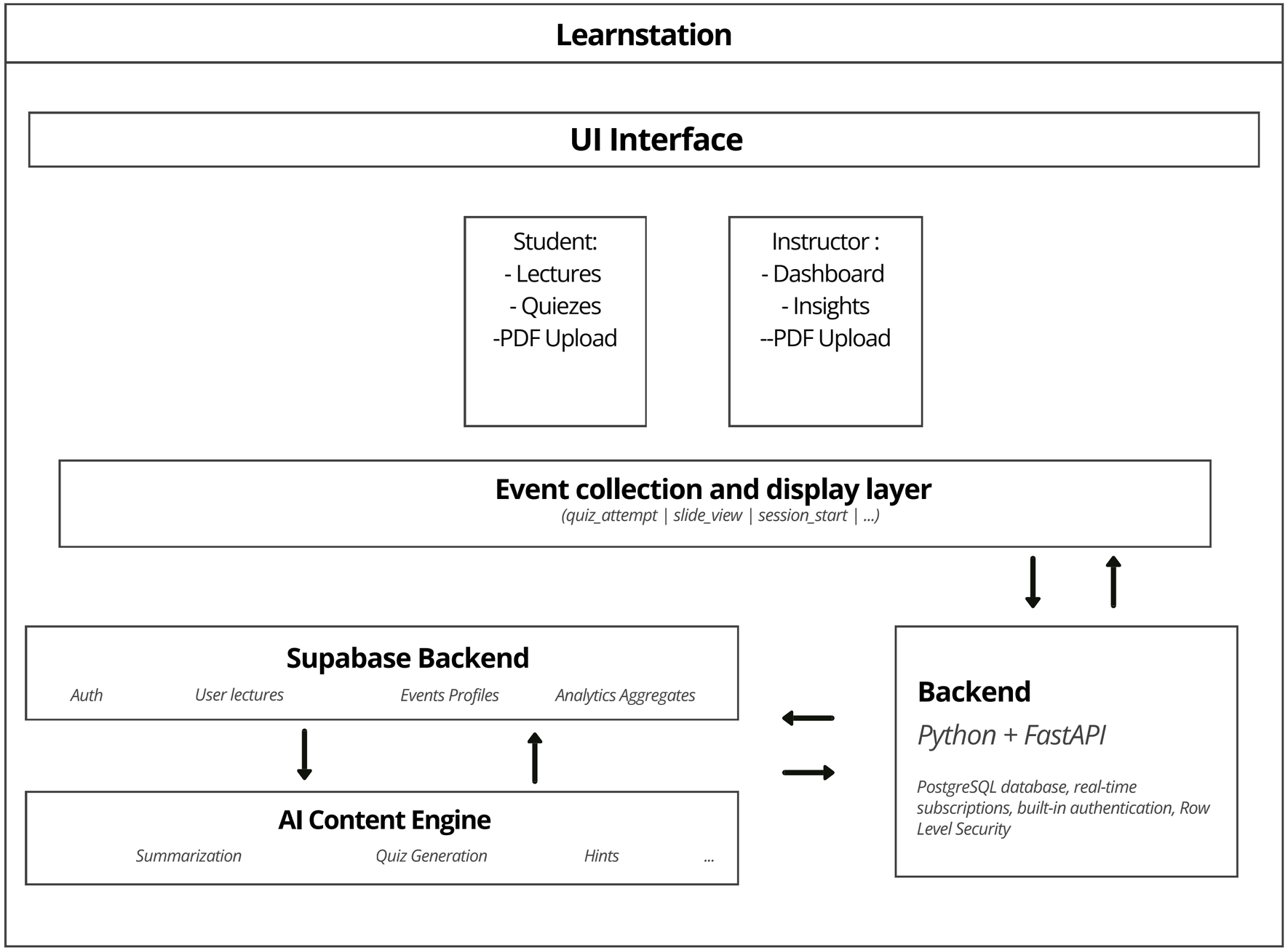
- Students/Instructor manually upload lecture materials (PDF)
- Prototyping without institutional dependencies
- Immediate evaluation capability

Phase B: ILIAS Integration Exploration

- Investigate ILIAS API capabilities for automated content synchronization



Components


Learnstation
UI Interface
Student:
- Lectures
- Quizes
-PDF Upload
Instructor :
- Dashboard
- Insights
--PDF Upload
Event collection and display layer
(quiz_attempt | slide_view | session_start | ...)
Supabase Backend
Auth
User lectures
Events Profiles
Analytics Aggregates
AI Content Engine
Summarization
Quiz Generation
Hints
...
Backend
Python + FastAPI
PostgreSQL database, real-time subscriptions, built-in authentication, Row Level Security

Technology Stack

Frontend

React + TypeScript

Styling

Tailwind CSS + Shadcn/ui

Backend

Python + FastAPI

AI Content Engine

Database

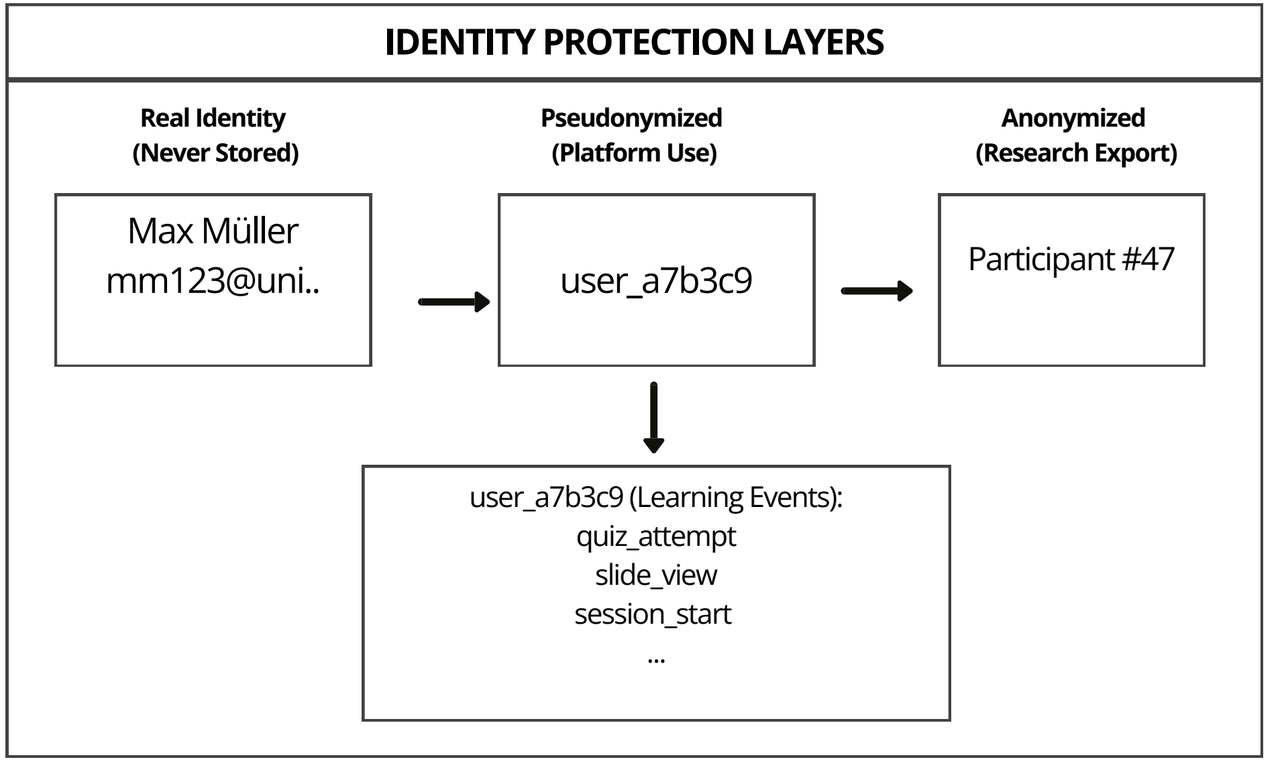
Self-Host Supabase

PostgreSQL database, real-time subscriptions, built-in authentication, Row Level Security

Student Data Protection (Datenschutz)

Data NOT Collected:

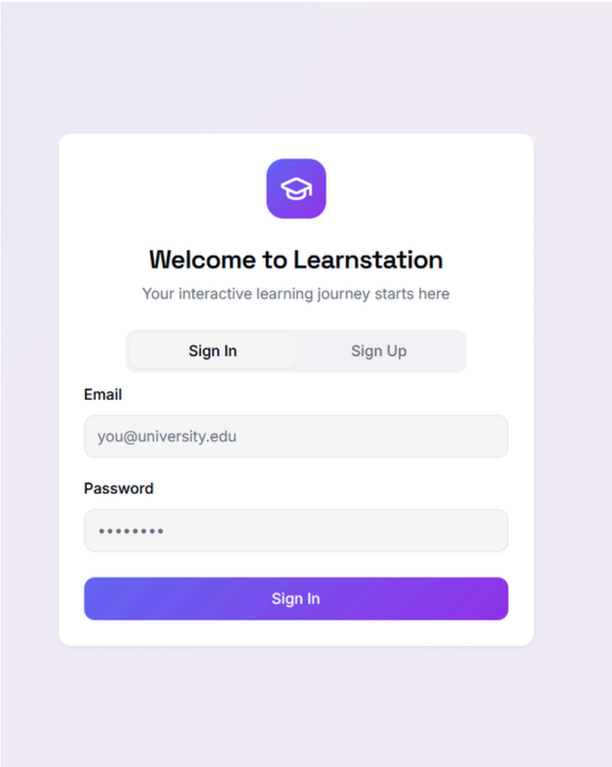
- Real names (unless explicitly provided and consented)
- Email addresses (Supabase Auth uses anonymized identifiers)
- Student ID numbers (Matrikelnummer)
- IP addresses (not logged)
- Precise geolocation
- Biometric data
- Health information
- Financial information

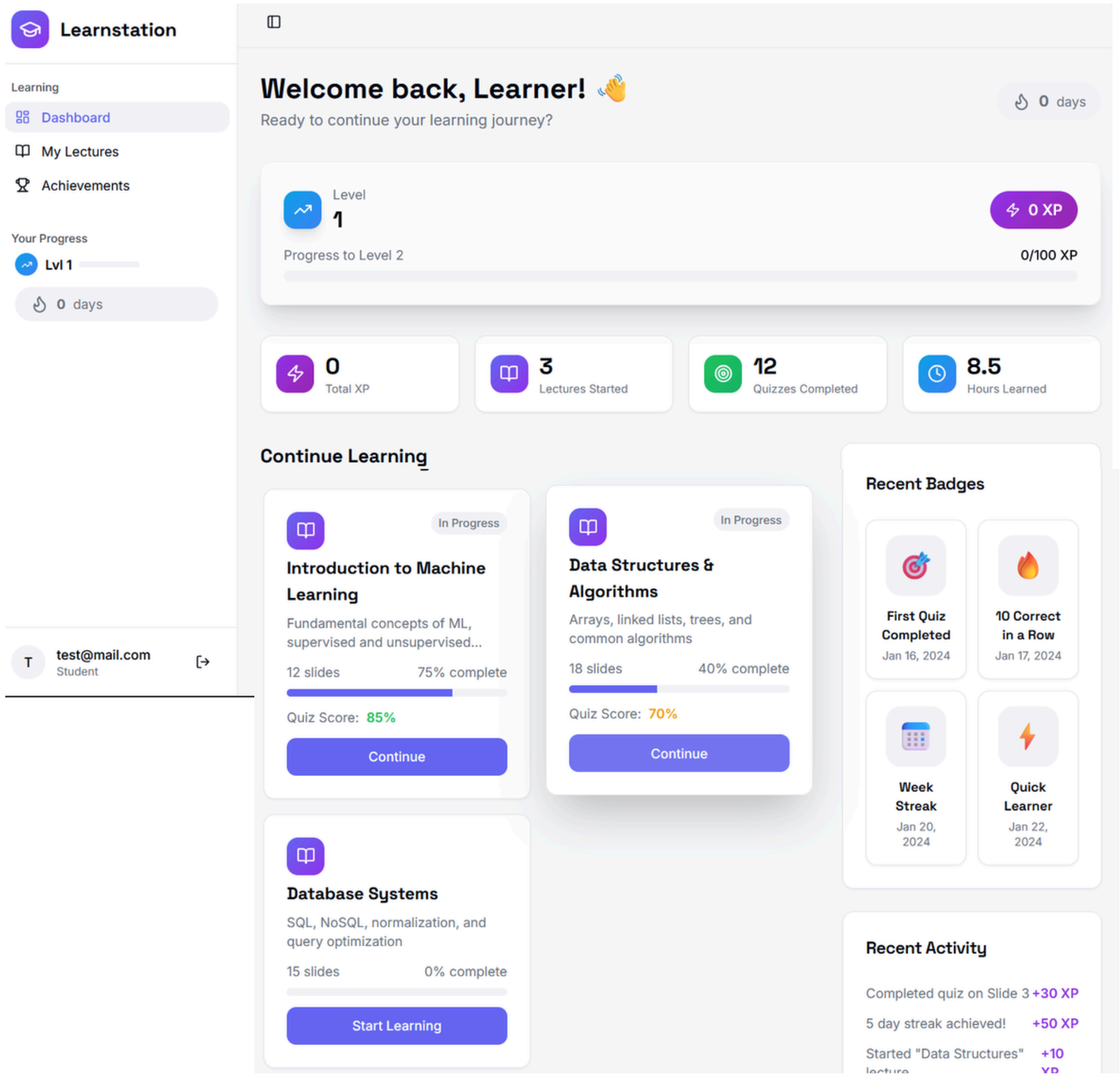


Data Collected (Pseudonymized):

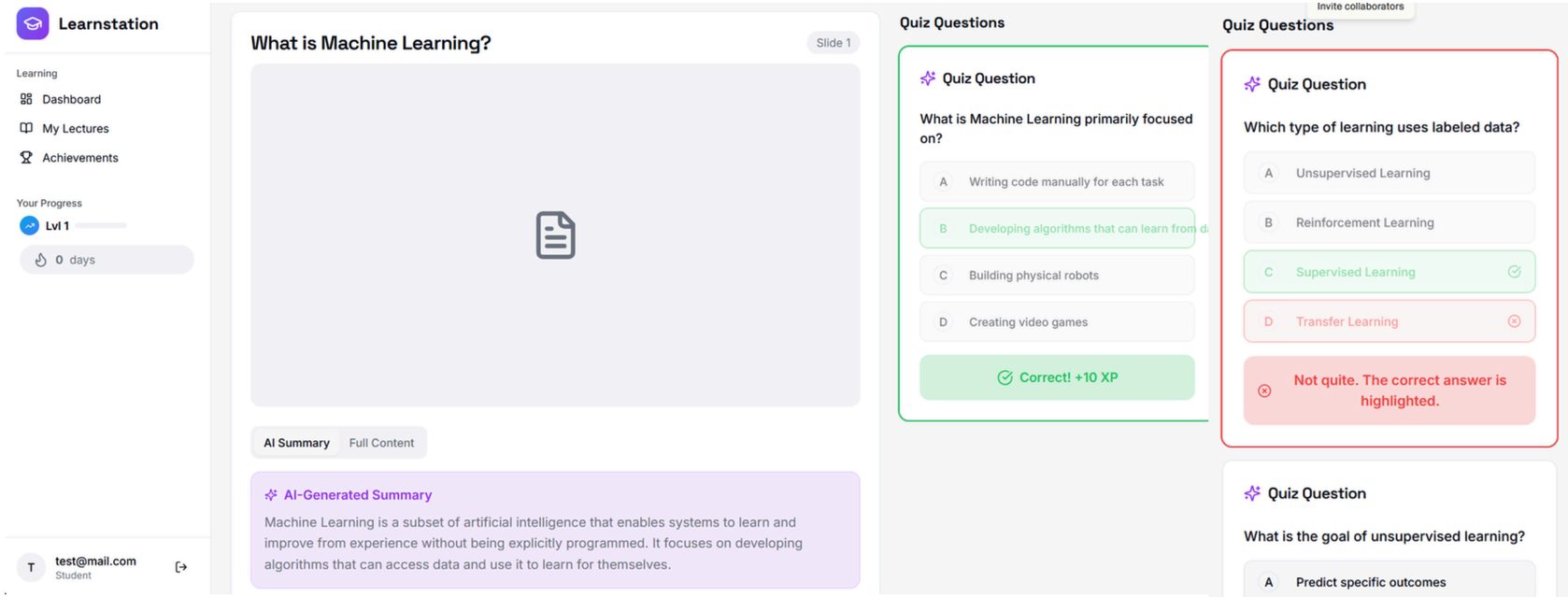
Data Category	Purpose
Pseudonymous user ID	Session continuity, progress tracking
Role (student/ instructor)	Access control, feature differentiation
Learning events (quiz attempts, slide views)	Analytics, research
Timestamps	Temporal pattern analysis
Performance scores	Progress tracking, gamification
Session metadata	Session metadata

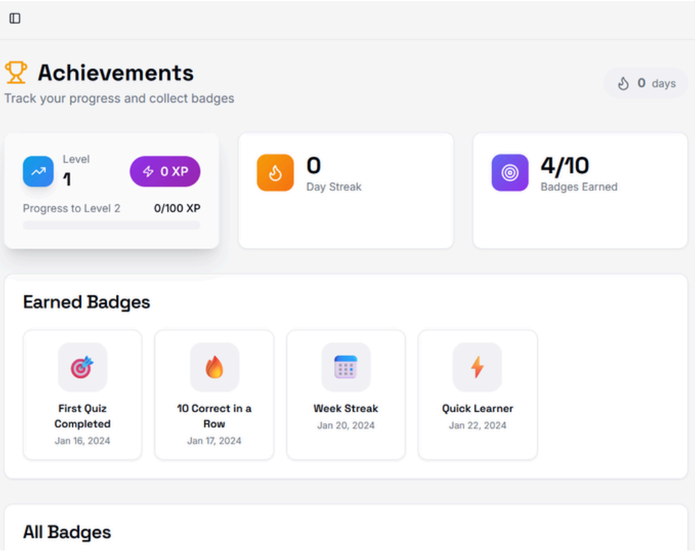
Visualization



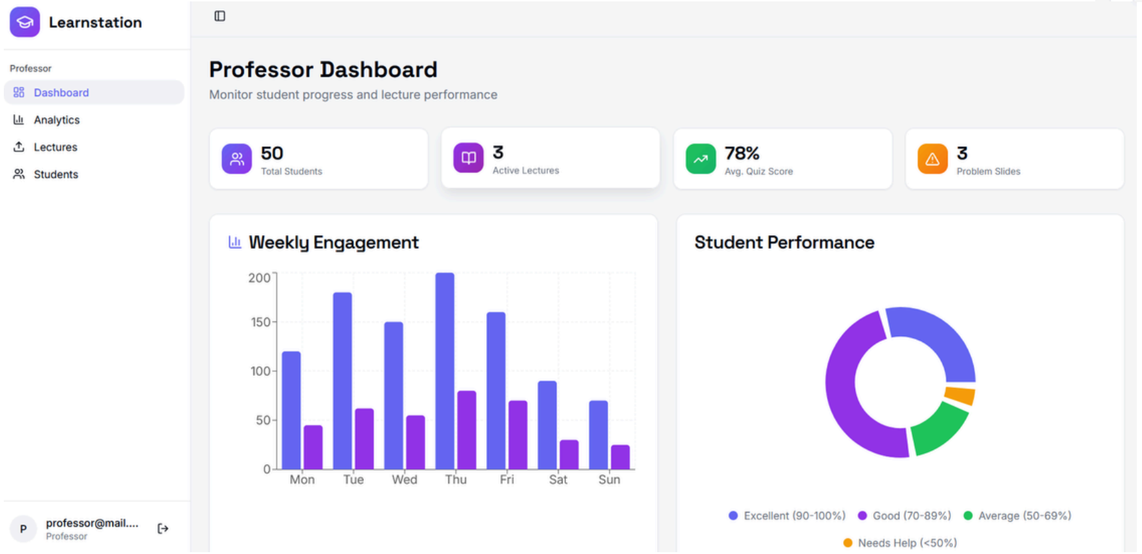


Student:





Instructor:



What Makes our Learnstation Different?

Feature	Our project	H5P	xAPI/LRS	ClassPoint	Quizizz
PDF upload & processing	✅	❌	❌	❌	❌
AI content generation	✅	❌	❌	✅	✅
Event-based data model	✅	Limited	✅	❌	Limited
Data profiling/analytics	✅	❌	✅	❌	Basic
Gamification (XP, levels)	✅	❌	❌	✅	✅

Thank You